Adding the contributions from the positive and the negative frequencies reconstructs the time domain signal:

$$\begin{array}{rl} \text{contribution from positive frequency} \rightarrow & \tfrac{1}{2}\cos(2\pi 0.23n) + \tfrac{1}{2}j\sin(2\pi 0.23n) \\ \text{contribution from negative frequency} \rightarrow & \tfrac{1}{2}\cos(2\pi 0.23n) - \tfrac{1}{2}j\sin(2\pi 0.23n) \\ \hline \text{resultant time domain signal} \rightarrow & \cos(2\pi 0.23n) \end{array}$$

In this same way, we can synthesize a sine wave in the time domain. In this case, we need a positive and negative frequency from the imaginary part of the frequency spectrum. This is shown by the round markers in Fig. 31-1. From Eq. 31-8, these spectral values contribute a sine wave and an imaginary cosine wave to the time domain. The imaginary cosine waves cancel, while the real sine waves add:

$$\begin{array}{rl} \text{contribution from positive frequency} \rightarrow & -\tfrac{1}{2}\sin(2\pi 0.23n) - \tfrac{1}{2}j\cos(2\pi 0.23n) \\ \text{contribution from negative frequency} \rightarrow & -\tfrac{1}{2}\sin(2\pi 0.23n) + \tfrac{1}{2}j\cos(2\pi 0.23n) \\ \hline \text{resultant time domain signal} \rightarrow & -\sin(2\pi 0.23n) \end{array}$$

Notice that a *negative* sine wave is generated, even though the positive frequency had a value that was *positive*. This sign inversion is an inherent part of the mathematics of the complex DFT. As you recall, this same sign inversion is commonly used in the real DFT. That is, a *positive* value in the imaginary part of the frequency spectrum corresponds to a *negative* sine wave. Most authors include this sign inversion in the definition of the real Fourier transform to make it consistent with its complex counterpart. The point is, this sign inversion *must* be used in the complex Fourier transform, but is merely an option in the real Fourier transform.

The *symmetry* of the complex Fourier transform is very important. As illustrated in Fig. 31-1, a real time domain signal corresponds to a frequency spectrum with an *even* real part, and an *odd* imaginary part. In other words, the negative and positive frequencies have the same sign in the real part (such as points 1 and 2 in Fig. 31-1), but opposite signs in the imaginary part (points 3 and 4).

This brings up another topic: the imaginary part of the time domain. Until now we have assumed that the time domain is completely real, that is, the imaginary part is zero. However, the complex Fourier transform does not require this.